

CCG-IDS: A Causal Counterfactual Graph-Based Intrusion Detection System for Industrial IoT

Chen Zhang[✉], Lu Zheng[✉], Huakun Huang[✉], *Member, IEEE*, and Chunhua Su[✉], *Member, IEEE*

Abstract—The industrial Internet of Things (IIoT) requires robust intrusion detection systems (IDS) to ensure critical business continuity. However, existing solutions suffer from high false-positive rates, difficulty in interpreting, and cross-domain generalization. In particular, they fail to reliably infer context-dependent causal links from host logs under global graph-level context. To address these issues, we propose causal counterfactual graph-based IDS (CCG-IDS), an IIoT-focused interpretable graph neural network IDS based on conformal calibration and counterfactual reasoning. This system implements a unified provenance sub-graph detection paradigm, and employs conformal anomaly detection and counterfactual reasoning to provide calibrated alerts and interpretable outputs. We also estimate predictive uncertainty via Fisher information to quantify decision confidence. We introduce a discrete counterfactual explainer with a counterfactual destructiveness score (CDS) to extract a minimal decision-critical evidence chain, and use this evidence to generate structured analyst-ready security reports. Experiments on real-world industrial log datasets, including the windows event log EVTX (Windows XML Event Log) and DARPA OpTC (Operationally Transparent Cyber) datasets, demonstrated that CCG-IDS achieved an *F1* score of 92% and a near-zero false-positive rate, outperforming other state-of-the-art methods.

Index Terms—Causal, counterfactual, graph, industrial Internet of Things (IIoT), intrusion detection system (IDS).

NOMENCLATURE

t	Time-Window index.
$\mathcal{G}_t = (V_t, E_t, X_t)$	Window provenance subgraph.
V_t, E_t, X_t	Nodes, edges, node features.
$N = V_t , L, d_h$	Nodes; GNN layers; hidden dim.
ρ	DropEdge probability.

Received 12 January 2026; accepted 19 February 2026. Date of publication 17 March 2026; date of current version 5 June 2026. This work was supported in part by the JSPS Grant-in-Aid for Scientific Research (C) under Grant 23K11103, in part by the National Natural Science Foundation of China under Grant 62571147, in part by the Natural Science Foundation of Guangdong Province, China under Grant 2025A1515011755, and in part by the SCAT Foundation under Grants for Researchers. Paper no. TII-26-0353. (*Corresponding author: Chunhua Su.*)

Lu Zheng is with the School of Computer and Communication, Lanzhou University of Technology, Lanzhou 730050, China.

Huakun Huang is with the School of Computer Science and Cyber Engineering, Guangzhou University, Guangzhou 510006, China.

Chen Zhang and Chunhua Su are with the Department of Computer Science and Engineering, University of Aizu, Aizuwakamatsu 965-8580, Japan (e-mail: chsu@u-aizu.ac.jp).

Digital Object Identifier 10.1109/TII.2026.3667569

1941-0050 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See <https://www.ieee.org/publications/rights/index.html> for more information.

$\mathbf{h}_v^{(l)}$	Node embedding at GNN layer l .
$\phi(e)$	Edge/event feature.
$\mathbf{z}_t$	Graph embedding (dual readout).
ℓ_t, p_t, s_t	Logits; attack probability; margin score.
w_0, w_1, π_0, π_1	CE weights; class priors (benign/attack).
$\alpha, \mathcal{N}_\alpha, \tau$	False-positive rate budget; hard negatives; rank smoothness.
λ_{rank}	Weight of ranking loss.
$\hat{p} = g(p)$	Calibrated probability (isotonic).
$U, \bar{U}, \beta, \lambda_u$	Fisher uncertainty; normalized; mixing; risk mix.
$\mathcal{N}_{\text{val}}, \theta_{\text{raw}}, \theta_w, \theta_{\text{ZF}}$	Val benign set; thresholds (raw/weighted/zero-FP).
w_t, ε, γ	Uncertainty-weighted quantile params.
S, K, T_f	Removed-edge set; budget; detector inference cost.
$\Delta_1(e), \text{CDS}(e), \omega_e$	Score drop; edge importance; CDS weight.
$\mathcal{G}^*, m_e$	Decision-critical subgraph; edge mask.

I. INTRODUCTION

THE Industrial Internet of Things (IIoT) environments face increasingly sophisticated host-based intrusions, such as advanced persistent threats (APTs) and lateral movement attacks [1]. These stealthy activities, exemplified by incidents like SolarWinds and NotPetya, are characterized by long-term inactivity and high stealth [2], [3]. In IIoT deployments, resource-constrained devices often lack robust security protections, making system audit logs one of the few reliable data sources for intrusion detection [4]. Host provenance logs record detailed system activity and causal relationships. Some research has demonstrated that by parsing this audit data into provenance graphs, representing processes, files, and network sockets as nodes and their interactions as directed edges, security systems can gain a holistic view of runtime behavior [5], [6]. This graph-based approach has significant implications for detecting APTs and zero-day attacks, where capturing the complete causal chain of an attack is important for early detection and mitigation [7]. However, deploying provenance-based host-based intrusion detection systems in real-world industrial environments still faces significant challenges.

1) *Sparse attack signals* [8]: APTs often generate only sparse, low-amplitude signals in host logs by blending in

with normal activity. An APT's provenance graph may contain only a few anomalous nodes or edges amidst thousands of benign events. This makes it difficult for detectors to distinguish malicious substructures from benign noise without generating numerous false positives.

- 2) *High false positives* [9], [10]: Highly sensitive IDSs often result in high false positives. The large number of legitimate operations similar to attacks, combined with missing logs, time asynchrony, and semantic heterogeneity, makes it difficult to establish stable baseline models and generalize poorly across domains, frequently misclassifying seemingly anomalous provenance substructures as malicious.
- 3) *Data silos* [11], [12]: Host log characteristics vary significantly across organizations, platforms, and over time. IDS models in one environment may not generalize directly to another.
- 4) *Lack of explainability* [13], [14]: Anomaly-based machine learning detectors can issue alerts without human-understandable justification. However, in mission-critical industrial environments, security operators need explanations for alert flags to verify genuine threats and facilitate incident response. Unfortunately, to date, most provenance-based IDS research has focused on detection accuracy rather than explainability. The decision-making process of graph neural networks (GNNs) and other models remains opaque, hindering analyst trust. This has prompted researchers to propose graph interpretability techniques to identify the most influential subgraph features on model predictions.

These challenges indicate that enterprises and IIoT deployments require an IDS that can achieve high detection, low false positives, adapt to environmental changes, and output interpretable results. In the past few years, researchers have proposed a variety of approaches, which we can roughly categorize as follows:

- 1) edge anomaly detection [15];
- 2) full-graph detection with attention mechanisms [16];
- 3) heterogeneous graph modeling and cross-domain learning [17];
- 4) interpretability techniques for graph-based IDS [18].

Despite these advances, no existing solution can fully meet the three requirements of low false positives, cross-source fusion, and interpretable output. This results in security analysts having to manually inspect large causal graphs to reconstruct attack steps, which is time-consuming and error-prone [19]. Given that current methods struggle to achieve low false positives, cross-source fusion, and interpretability, this article proposes a unified approach for collaborative optimization. To address these limitations, we propose CCG-IDS, an end-to-end intrusion detection pipeline built on unified provenance graphs derived from Windows host audit logs. CCG-IDS encodes each time-window graph with a GNN and summarizes it using dual readout pooling to capture both the overall behavior and the most suspicious evidence. It then uses Fisher uncertainty to estimate confidence in the prediction and to support low false-positive rate (FPR) operation. For interpretability, CCG-IDS generates a counterfactual explanation by removing the smallest set of decision-critical edges that makes the risk score drop below

the alert threshold, and reconstructs the corresponding causal path, which is quantified by the counterfactual destructiveness score (CDS). Finally, an LLM (Large Language Model)-based structured reporting module turns the extracted evidence into a fixed-schema, analyst-facing security report. Although prior provenance-based IDSs often employ GNN encoders, they solve different operational problems. ThreatTrace [6] represents the node-level role/type deviation paradigm, where anomalies are detected as entities whose learned benign roles deviate at runtime. FLASH [20] represents a comprehensive representation-learning and scalable deployment paradigm, enriching node semantics and employing lightweight scoring with downstream aggregation such as attack-evolution graphs. Both methods operate on provenance graphs and use GNN-based modeling, making them the most relevant references for positioning our contributions beyond other GNNs for provenance. Since provenance-graph IDS works may appear similar at a high level, we explicitly align the three frameworks along checkable axes, such as decision unit, false-positive control, and explainability paradigm. Table I summarizes these differences, showing that our framework changes the detection semantics from node alerts to time-windowed subgraph decisions and introduces a calibrated low/zero-FP (low/zero-False Positive) operating layer with counterfactual intervention-based evidence. To avoid treating the above differences as qualitative claims, we provide a Table II that summarizes our contributions and solutions compared to these two baselines.

Our contributions are as follows.

- 1) We formulate intrusion detection on time-windowed provenance subgraphs and propose a GNN detector with dual readout pooling to robustly aggregate graph representations under large and variable graph sizes.
- 2) We introduce Fisher-information-based uncertainty to derive an uncertainty-aware risk score and apply conformal-style calibration to support two deployment modes that are zero-FP and low-FPR.
- 3) We propose a discrete counterfactual explainer that identifies a small decision-critical edge set whose removal reduces the risk score below the alert threshold; we rank evidence with the CDS and reconstruct a compact causal event chain for interpretation.
- 4) We design a constrained LLM module that converts the grounded counterfactual evidence into a fixed-schema JSON (JavaScript Object Notation) security report for analyst-facing triage.
- 5) We evaluate CCG-IDS on EVTX and DARPA OpTC and report both threshold-free ranking metrics and low-FPR operating-point metrics, together with efficiency measurements for deployment.

The rest of this article is organized as follows. Section II reviews related work. Section III presents our proposed framework and mathematical models, and Section IV evaluates our proposed performance. Finally, Section V concludes this article.

II. RELATED WORK

A. Node/Entity-Level Anomaly Detection

Node-level anomaly detection considers each process, file, or socket in the audit log graph as a potential compromise point

TABLE I
METHODOLOGICAL COMPARISON WITH OTHERS

Axis	ThreatTrace	FLASH	CCG-IDS (Ours)
Decision unit	Node/entity	Node (scores) + aggregation	Time-windowed subgraph G_t
Learning target	Benign role / type consistency	Representation quality + scalability	Low-FPR detection + calibrated operation
Low/zero-FP control	Heuristics (waiting/tolerance)	Post-hoc thresholding / ranking	Uncertainty-aware conformal modes
Explanation paradigm	Node-centric tracing	Aggregated evolution graphs	Counterfactual necessity (minimal edge removals)
Evidence format	Anomalous entities	Alert ranking + AEG context	Minimal causal chain + structured report
Cross-domain evaluation	Host provenance datasets	Multiple datasets	EVTX + OpTC (heterogeneous IIoT logs)

TABLE II
CONTRIBUTION SUMMARY AND THE SPECIFIC GAPS RELATIVE TO NODE-LEVEL PROVENANCE IDS BASELINES

Contribution	Solutions
Time-windowed subgraph decision G_t	Avoids fragmented node alerts; matches incident-triage semantics; captures distributed weak signals
Uncertainty-aware conformal calibration	Turns low-FPR / zero-FP operation into a method component, not a post-hoc threshold heuristic
Counterfactual perturbation + CDS ranking	Provides causal necessity evidence beyond importance-based explanations

and learns embeddings or features of these nodes to identify outliers [21]. Hassan et al.’s UNICORN [22] is an anomaly-based APT detector that explores the behavior of each node in a large provenance graph. By tailoring its modeling to the unique characteristics of APTs, UNICORN can detect hidden anomalies in a group of network hosts without relying on known attack signatures. Their evaluation shows that UNICORN outperforms previous state-of-the-art techniques and detects real-world APT scenarios with high accuracy. Wang et al.’s “You Are What You Do” [23] applied data provenance analysis to search for stealthy malware, implicitly learning normal patterns in system call graphs and flagging inconsistent node behavior. Wang et al.’s ThreatTrace [6] trains a GraphSAGE-based model by assigning each benign node’s label as its node type and extracting edge-type count distributions as node features. Runtime anomalies are nodes that become misclassified, with alert suppression relying on waiting-time/tolerance heuristics in streaming execution. During detection, ThreatTrace identifies nodes that significantly deviate from these learned benign roles as potentially malicious. This approach of modeling fine-grained entity roles has proven effective. Our formulation differs in the supervised signal and decision unit that we classify time-windowed provenance subgraphs G_t as benign/attack, which directly matches incident triage and enables window-level false-positive control and counterfactual causal-chain extraction.

B. Full-Graph Detection With Attention Mechanisms

To detect more complex multistage attacks, researchers have proposed methods that use event subgraphs or even the global graph as units of analysis [24]. These methods focus on contextual and structural patterns across entities, addressing the shortcomings of isolated node-level review that often miss such patterns. A representative work is ProGrapher [25], an anomaly detection system that operates on provenance graph snapshots. ProGrapher addresses the “dependency explosion” problem by

partitioning a continuous event stream into manageable time-windowed subgraphs. Each snapshot graph preserves the temporal order of events and is treated as an independent graph instance for detection.

Another notable subgraph-level approach is FLASH [20], which emphasizes comprehensive graph representation learning to improve accuracy and scalability. FLASH uses a Word2Vec-based semantic encoder to embed attributes, such as process names, command-line arguments, and file paths, capturing textual and categorical semantics often overlooked by earlier PIDSs. Second, it employs a GNN-based context encoder to encode the local and global structural context of each node in the provenance graph. By combining these, FLASH generates rich node embeddings that explain what each entity is and what it does in the graph. While these embeddings scale better to big data environments than naive node-by-node analysis, interpretability along causal chains remains insufficient, preventing security analysts from identifying counterfactual logic. Our contribution is complementary in focus, as we focus on deployment-facing low/zero-FP control through uncertainty-aware conformal calibration, and on counterfactual necessity-based explanations that produce minimal causal evidence chains for analyst verification.

C. Interpretability of Graph-Based IDSs

With the increasing sophistication of IDSs, interpretability has become a primary concern, particularly for analysts’ trust and verification of warnings. GNN interpretability transforms black-box scores into verifiable evidence by identifying the key subgraphs that best support current predictions. CF-GNNExplainer [26] searches for the smallest perturbation to the input graph that changes the model’s predictions. Deleting the resulting set of edges flips the results. Similarly, the framework of CF-2 [27] formalizes the dual goals of necessary and sufficient conditions for GNN explainability: an interpretable subgraph

TABLE III
EVENT 5-TUPLE SCHEMA $e = \{\text{SUB}, \text{ACT}, \text{OBJ}, \tau, \psi\}$

Field	Meaning	Type / Range	Example
sub	Subject entity (actor)	Entity $\in \{P, F, R, U, H, K\}$	$P : \text{powershell.exe}@host1...$
act	Action type between sub and obj.	Categorical label (finite set)	proc_create, file_read, file_write...
obj	Object entity (target).	Entity $\in \{P, F, R, U, H, K\}$	$P : \text{cmd.exe}...$
τ	Event time inside the window.	Timestamp or integer bucket	$\tau = 2025-12-24 \ 10:00:07$
ψ	Event Source.	Hostname	host = workstation-07

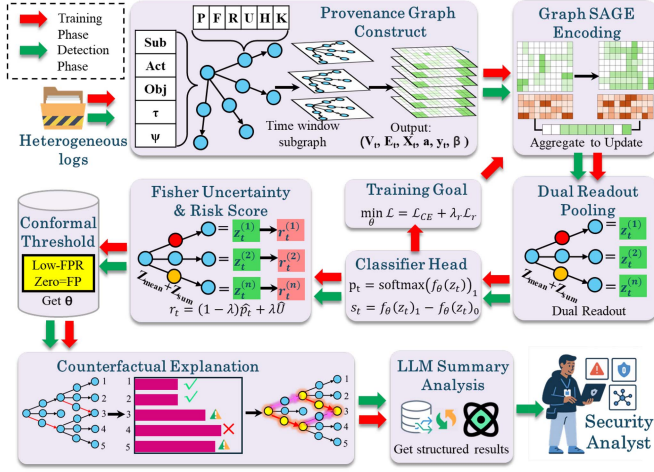


Fig. 1. CCG-IDS architecture.

that satisfies both factual and counterfactual conditions, while also keeping the explanation as simple as possible.

The fields of graph-based intrusion detection and explainable machine learning are converging towards solutions that not only detect APTs with low false positives but also output explainable decisions for decision makers. CCG-IDS builds on previous work to establish a unified framework for causal, conformal, graph-based host intrusion detection in IIoT environments.

III. METHODOLOGY

CCG-IDS is an end-to-end, explainable intrusion detection framework for IIoT environments. Fig. 1 illustrates the CCG-IDS framework. It first normalizes heterogeneous logs into event 5-tuples $\{\text{sub}, \text{act}, \text{obj}, \tau, \psi\}$, as shown in Table III, where sub is the subject entity, the actor that triggers the event, obj is the object entity, the target affected by the event, act is the action/relation type, τ is the event time or a discretized time-bucket within a window, and ψ is which host this event belongs to. Within each fixed time window t , we perform entity canonicalization and relation extraction to construct a directed provenance subgraph $G_t = (V_t, E_t, X_t, A_t)$. Nodes V_t correspond to canonical entities, including process (P), file (F), remote/socket (R), user (U), host (H), and registry (K), with node features X_t composed of type embeddings, structural statistics, and optional semantic cues. Directed edges E_t encode $\text{sub} \rightarrow \text{obj}$ interactions and are associated with edge attributes A_t , e.g., action one-hot, relative

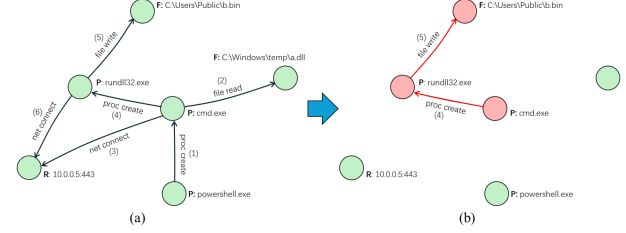


Fig. 2. Provenance subgraph demo.

sub	act	obj	τ	ψ
powershell.exe	proc create	cmd.exe	2024-12-24 10:00:07	workstation-07
cmd.exe	file read	C:\Windows\temp*.dll	2024-12-24 10:02:31	workstation-07
cmd.exe	net connect	10.0.0.5:443	2024-12-24 10:03:15	workstation-07
cmd.exe	proc create	rundll32.exe	2024-12-24 10:06:11	workstation-07
rundll32.exe	file write	C:\Users\Public\b.bin	2024-12-24 10:07:08	workstation-07
rundll32.exe	net connect	10.0.0.5:443	2024-12-24 10:11:03	workstation-07

Fig. 3. Actual provenance subgraph data example.

time, counts/bytes, and ports. Repeated events within a window are merged into weighted edges. The subgraph is encoded by a multilayer GraphSAGE, which aggregates neighbor information to produce node representations $H^{(L)} = \mathbf{h}_v^{(L)}$. Fig. 2 visualizes our provenance modeling: within each time window t , we construct G_t by canonicalizing entities and converting each event into a directed interaction edge. This graph representation preserves the operational dependencies among processes/files/network endpoints and enables both learning-based detection and post hoc evidence extraction. The actual provenance subgraph data is shown in Fig. 3. Our detector predicts a risk score for the entire window-level subgraph G_t . This differs from node-level detectors that assign anomaly scores to entities and later aggregate alerts. Window-level decisions align with operational triage that analysts typically investigate bursts of activity in time and demand an evidence chain that explains why the window is suspicious. Window-level classification enables explicit low/zero-FP operating-point control over alert streams and counterfactual interventions that remove a minimal set of causal edges to flip the window-level decision, yielding necessary evidence chains.

We then use dual readout pooling with two parallel readouts: $z_{\text{mean}} = \frac{1}{N} \sum_v \mathbf{h}_v^{(L)}$ and $z_{\text{sum}} = \frac{1}{\sqrt{N}} \sum_v \mathbf{h}_v^{(L)}$, are concatenated into the graph vector $\mathbf{z}_t = [z_{\text{mean}} | z_{\text{sum}}]$, which feeds a small MLP head to produce logits, probability p_t , and margin s_t . Training

jointly optimizes class-weighted cross-entropy and a low-FPR ranking loss that pairs positives only with the top- α hardest negatives, directly improving true positive rate (TPR) at low FPR. On validation, we calibrate probabilities with isotonic regression to obtain $\hat{p}_t = g(p_t)$ and compute a Fisher-aware uncertainty score via gradient-norm proxies on the head and on $\mathbf{z}_t$, which is normalized to $\tilde{U}$. These are fused into a risk score r , so uncertain samples are treated more conservatively. For decisions, we set a conformal threshold once on benign validation windows with a weighted quantile (low-FPR mode) that bounds the online FPR, or the benign maximum (zero-FP mode) for critical assets. Online inference is then $\mathcal{O}(1)$: compute r_t and trigger an alert if $r_t \geq \theta$. Upon alert, we provide causal transparency with a counterfactual explainer that searches for a minimal edge set $S \subseteq E_t$ whose removal flips the decision below the operating threshold, and rank evidence by a CDS $\text{CDS}(e) = \hat{p}_1(G_t) - \hat{p}_1(G_t \setminus e)$.

The resulting key path G_t^* (minimal causal evidence chain) plus context is passed to a fine-tuned LLM to produce a fixed-schema structured security summary in JSON, including summary, entities, timeline, indicators, suspected stages, and recommended actions.

The Nomenclature shows the meaning of some formula symbols. where $\oplus$ denotes optional edge-feature fusion, AGG is mean, and σ is rectified linear unit (ReLU) with dropout. During training, we apply DropEdge with probability ρ .

A. Dual Readout Pooling

For variable-sized subgraphs, sum readout captures counting-like signals but scales with $|V_t|$, whereas the mean readout is stable but may dilute rare signals. We compute both in parallel

$$\mathbf{z}_{\text{mean}} = \frac{1}{N} \sum_{v \in V_t} \mathbf{h}_v^{(L)} \quad (1)$$

$$\mathbf{z}_{\text{sum}} = \frac{1}{\sqrt{N}} \sum_{v \in V_t} \mathbf{h}_v^{(L)}, \quad N = |V_t| \quad (2)$$

and concatenate them: $\mathbf{z}_t = [\mathbf{z}_{\text{mean}} | \mathbf{z}_{\text{sum}}]$. The scaling factor $1/\sqrt{N}$ keeps the expected magnitude of $\mathbf{z}_{\text{sum}}$ approximately constant under the central limit theorem, preventing gradient explosion on large graphs. Together with $\mathbf{z}_{\text{mean}}$, it balances robustness to scale and sensitivity to rare events. A classifier head $f_\theta : \mathbb{R}^{2d_h} \rightarrow \mathbb{R}^2$ produces logits ℓ_t . The attack probability and score are $p_t = \text{softmax}(\ell_t)_1$, and $s_t = \ell_{t,1} - \ell_{t,0}$. Under severe class imbalance, we combine a class-weighted cross-entropy with a pAUC-oriented ranking loss that emphasizes the low-FPR regime. The cross-entropy is

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B} \sum_{t \in \mathcal{B}} (w_1 y_t \log p_t + w_0 (1 - y_t) \log (1 - p_t)) \quad (3)$$

where $w_1 \propto 1/\pi_1$ and $w_0 \propto 1/\pi_0$ depend on class frequencies. To emphasize negatives most likely to cause false alarms, let $\mathcal{P} = \{i : y_i = 1\}$ and $\mathcal{N} = \{j : y_j = 0\}$. Take top- α quantile

hard negatives $\mathcal{N}_\alpha \subset \mathcal{N}$, and define

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{P}| |\mathcal{N}_\alpha|} \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{N}_\alpha} \log(1 + \exp(-(s_i - s_j)/T)) \quad (4)$$

where $T > 0$ controls smoothness. The joint objective is

$$\min_{\theta} \mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}. \quad (5)$$

This focuses learning on the low-FPR frontier, directly improving $\text{TPR@FPR} \leq \alpha$ and $\text{pAUC@}\alpha$. To obtain comparable and well-calibrated scores, we use isotonic regression to learn a monotone calibration map $g(\cdot)$ on the validation set

$$\hat{p} = g(p) = \arg \min_{g \in \mathcal{M}} \sum_{t \in \mathcal{V}} (y_t - g(p_t))^2 \quad (6)$$

where $\mathcal{M}$ is the set of nondecreasing functions. This ensures $\hat{p}$ approximates true posterior probabilities for consistent thresholds.

B. Fisher Uncertainty

Let the predicted class be $c = \arg \max_k \text{softmax}(\ell)_k$ and define the negative log-likelihood $\mathcal{J} = -\log \text{softmax}(\ell)_c$. We approximate empirical Fisher sensitivities in the head-parameter space and representation space

$$U^{\text{head}} = \|\nabla_{\theta_{\text{head}}} \mathcal{J}\|_2^2 \quad (7)$$

$$U^{\text{feat}} = \|\nabla_{\mathbf{z}} \mathcal{J}\|_2^2 \quad (8)$$

$$U = U^{\text{head}} + \beta U^{\text{feat}}. \quad (9)$$

After robust normalization to $\tilde{U} \in [0, 1)$, we define a risk score that interpolates calibrated probability and uncertainty

$$r = (1 - \lambda_u) \hat{p} + \lambda_u \tilde{U}, \quad \lambda_u \in [0, 1]. \quad (10)$$

Larger $\tilde{U}$ implies lower model confidence; mixing $\tilde{U}$ into r yields conservative decisions when needed.

C. Conformal Thresholding With Uncertainty Awareness

On validation-time benign samples $\mathcal{N}_{\text{val}}$, we compute risk scores $\{r_t\}$ and set a distribution-free $(1 - \alpha)$ quantile

$$\theta_{\text{raw}} = \text{Quantile}_{1-\alpha}(\{r_t : t \in \mathcal{N}_{\text{val}}\}). \quad (11)$$

To be more robust under domain shift, we use uncertainty-weighted quantiles with weights $w_t \propto (\varepsilon + \tilde{U}_t)^\gamma$ ($\gamma > 1$)

$$\theta_w = \text{WQuantile}_{1-\alpha}(\{r_t\}, \{w_t\}). \quad (12)$$

For ultraconservative operation, we deploy a zero-false-positive threshold $\theta_{\text{ZF}} = \max_{t \in \mathcal{N}_{\text{val}}} r_t$. These thresholds support two operational modes: Low-FPR (e.g., $\alpha = 1\%$ or 5%) and zero-FP.

D. Counterfactual Explanation and Path Reconstruction

Given an alerted subgraph $\mathcal{G} = (V, E)$ with attack score $\hat{p}_1 = \hat{p}(y=1)$, we seek the smallest edge set whose removal flips the decision or drops below a normal threshold θ

$$\min_{S \subseteq E} |S| \quad \text{s.t.} \quad \hat{p}_1(\mathcal{G} \setminus S) < \theta. \quad (13)$$

This is NP-hard, so we apply a discrete greedy approximation. For each edge $e \in E$, define the instantaneous destruction gain

$$\Delta_1(e) = \hat{p}_1(\mathcal{G}) - \hat{p}_1(\mathcal{G} \setminus \{e\}) \quad (14)$$

and remove edges in descending Δ_1 order until the constraint holds or a budget K is reached, yielding $\hat{\mathcal{S}}$. We quantify contribution via the CDS

$$\text{CDS}(e) = \Delta_1(e) \quad (15)$$

$$\text{CDS}(v) = \sum_{e \in E: e \text{ incident to } v} \omega_e \text{CDS}(e) \quad (16)$$

where ω_e can be set according to edge type or temporal weight to make different events comparable in importance. $\text{CDS}(v)$ is used to highlight key processes or hosts, and $\text{CDS}(e)$ to highlight critical event chains. By taking the connected closure of the top- k edges ranked by CDS, we obtain the attack-path subgraph $\mathcal{G}^*$. And we introduce differentiable masks $m_e \in [0, 1]$ to reduce the number of forward passes, multiplying each edge message by m_e , and optimizing

$$\min_{m \in [0, 1]^{|E|}} \hat{p}_1(\mathcal{G}; m) + \lambda_1 |m|_1 \quad \text{s.t. } |m|_0 \leq K \quad (17)$$

using projected gradient. Finally, we binarize m to obtain $\hat{\mathcal{S}} = \{e : m_e = 0\}$. We feed $\mathcal{G}^*$ (entities, edges, timestamps, endpoints, file paths) and CDS rankings to a fine-tuned large language model with a constrained prompt to produce a fixed-schema JSON report.

Let K be the maximum number of edges to remove and T_f be the inference cost of the detector on a window graph G . The greedy counterfactual search performs at most K iterations, and each iteration requires a bounded number of forward evaluations to test candidate edge removals and check whether the score falls below the operating threshold. Therefore, the overall time complexity is $O(K \cdot T_f)$.

E. LLM-Based Structured Reporting

The LLM module is a post hoc reporting component that translates already-computed detector outputs and counterfactual evidence into an analyst-facing JSON report. It does not affect detection, calibration, or alert triggering.

Inputs: The LLM receives only structured facts extracted from the detector and the counterfactual explainer, including: window metadata (duration, host ID), calibrated risk score and decision, counterfactual summary statistics (e.g., number of removed edges, keep ratio), and a ranked evidence list (top CDS edges with endpoints and timestamps). Raw logs and external context are not provided.

Output: The LLM outputs a fixed-schema JSON object: {summary, key_indicators, risk_level, action_items}, where $\text{risk_level} \in \{\text{low, medium, high}\}$. We enforce JSON-only output under schema validation.

Prompt template: We use a two-part prompt: 1) a system instruction that specifies the strict JSON schema; 2) a user prompt that contains only the input facts and explicitly forbids introducing unseen entities/events. Unknown details must be returned as “unknown.”

Hallucination safeguards: We constrain the model to a closed world of provided facts, enforce JSON-only output, and validate outputs via JSON parsing and schema checks. Malformed outputs are rejected and replaced with a safe minimal report (explicit “unknown” fields). We additionally flag any report item that is not grounded in the provided evidence list as unsupported and exclude it from operational decision-making.

IV. EXPERIMENTS

To evaluate the effectiveness of CCG-IDS, we conducted experiments on various datasets, including the Windows event log dataset EVTX and the DARPA-provided OpTC dataset. The experiments covered multiple aspects, including detection accuracy, false-positive control, and explanation performance, and compared our approach with existing methods.

A. Setup

*EVTX dataset:*¹ Security logs from Windows hosts in industrial control environments. The EVTX files were converted to XML/JSON to extract time, event ID, process name, and other information. These logs were then mapped into a time-series attribution graph, segmented into fixed 1-min windows, and annotated based on the start and end times of known attacks.

*OpTC dataset:*² DARPA’s large-scale APT simulation (approximately 1000 hosts, 500 collected, and run for two weeks), with benign early-stage and red team injections later. We used host-side eCAR 25 September events, approximately 15 GB of data, with distributed parsing, constructed using sliding 1-min windows, and annotated with official ground truth.

EVTX and OpTC differ substantially in event sources, schema richness, and graph scale/density. To test domain robustness, we keep the same normalization (5-tuple), windowing policy, and model configuration across both datasets, and report both ranking metrics and low-FPR operating-point behavior. Consistent performance across these heterogeneous pipelines supports that the framework is not specialized to one dataset’s logging idiosyncrasies.

Data preprocessing: On Linux, L40S GPU RAM 48 G, Python 3.11, we use python-evtx/lxml and ijson to parse EVTX and OpTC eCAR. We unify raw observability into edges {sub, act, obj, τ , ψ }, deduplicate to nodes, and apply 60 s windowing to produce per-window edges and windows. EVTX windows are labeled via host prefix atk/ben, while OpTC uses GT/IoC. We export the PyTorch Geometric graphs to pt files and perform a temporal split of 70/15/15. The training set, 70%, is used to train our subgraph detection method, and we also use this training portion to fine-tune Qwen-32B. The validation and test sets each account for 15%.

Evaluation metrics.

- 1) *AUPR:* Area under the precision–recall curve, preferred for imbalanced data. Computed over detection scores on all test subgraphs.

¹[Online]. Available: <https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES>

²[Online]. Available: <https://github.com/FiveDirections/OpTC-data>

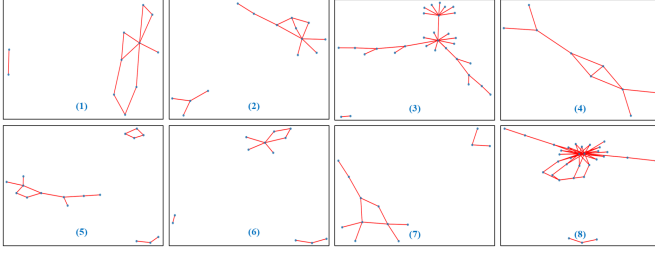


Fig. 4. Representative high-confidence alerts and counterfactual analyses from the EVTX dataset.

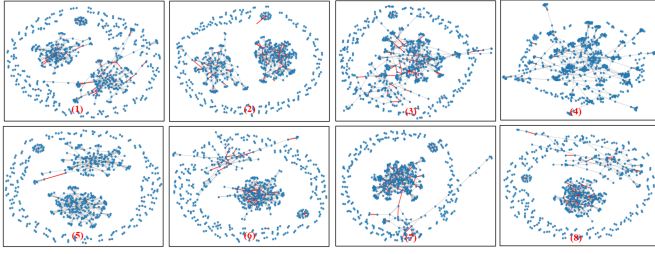


Fig. 5. Representative high-confidence alerts and counterfactual analyses from the OpTC dataset.

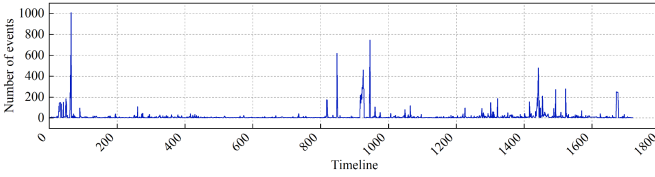


Fig. 6. Temporal distribution of event occurrences in the EVTX datasets.

- 2) $\text{TPR@FPR} = x\%$: TPR at fixed FPRs. We emphasize 1% and 5%. Captures recall under strict false-alarm budgets, which is critical for industrial use.
- 3) Precision@K : Precision among the top- K alerts ($K = 50/100/200$). Reflects ranking quality and how many of the most urgent alerts are truly malicious, reducing analyst load.
- 4) $\text{pAUC@}\alpha$: Normalized AUC over $\text{FPR} \in [0, \alpha]$.
- 5) Other metrics: We also report receiver operating characteristic – area under the curve (ROC-AUC) and $F1$ score. Explanation quality is assessed via case studies and an auxiliary explanation hit rate.

Comparison methods: To clarify the methodological novelty and ensure a fair comparison, we select two representative provenance-graph intrusion detection baselines: *ThreatTrace* and *FLASH*.

B. Experimental Results

Figs. 4–6(a) and 6(b) illustrate the temporal distribution of event counts in the EVTX and OpTC datasets. As shown, the OpTC dataset exhibits a substantially higher number of events

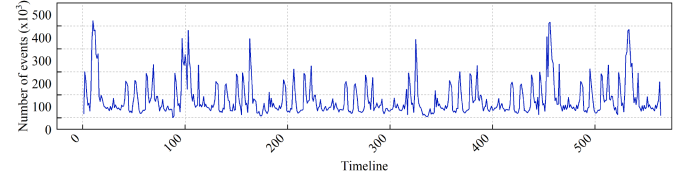


Fig. 7. Temporal distribution of event occurrences in the OpTC datasets.

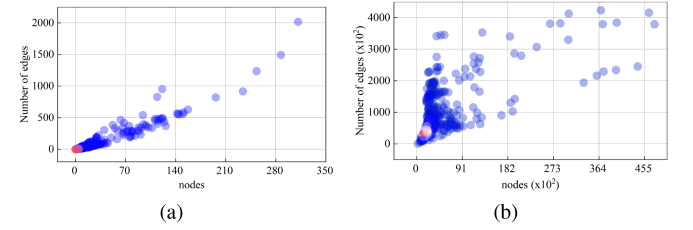


Fig. 8. Event distributions across different node scales in the EVTX and OpTC dataset. (a) EVTX datasets; (b) OpTC dataset.

within the same time span, along with a larger set of participating nodes. Accordingly, we first conduct the full experimental pipeline on the smaller EVTX dataset and then evaluate the robustness of our approach on the more complex OpTC dataset.

Figs. 7 and 8 present 16 representative alerts from the EVTX and OpTC datasets, respectively. We first score and rank all test-set subgraphs with CCG-IDS. Then, under the Zero-FP operating mode, we derive a decision threshold on the validation set via Fisher uncertainty aware conformal and retain test samples with score $\geq$ threshold. From these candidates, we select the Top 16 for display. For each selected subgraph, we run a discrete counterfactual search: edges are removed in decreasing CDS order until the model's score falls below the threshold or the search budget is reached. For visualization, the original graph and the minimal counterfactual graph are overlaid, highlighting critical edges in red. Finally, we submit the key events and context to fine-tuned LLM in JSON mode, producing a structured counterfactual summary and recommended actions, as shown in Fig. 9. These figures demonstrate the causal critical paths revealed by minimal counterfactual perturbations, and analyst-ready structured summaries that accelerate triage and root-cause analysis.

Fig. 10 achieves strong ranking performance on the EVTX dataset, while its AUPR of 0.9218 is slightly lower than the AUC, indicating a small overlap between positive and negative scores in the very low recall region. Fig. 11 shows that CCG-IDS also performs well overall on the OpTC dataset, achieving an AUC of 0.9641 and an AUPR of 0.9698, with both values being nearly identical. This demonstrates that the model maintains stable performance across the entire range from high precision to high recall.

Fig. 12 presents the normalized partial ROC area at $\text{FPR} \leq 10\%$. On EVTX, we obtain $\text{pAUC@10\%} = 0.8723$, and on OpTC, we obtain 0.8372. Because normalized $\text{pAUC@10\%}$ equals the average TPR within 0%–10% FPR, these values

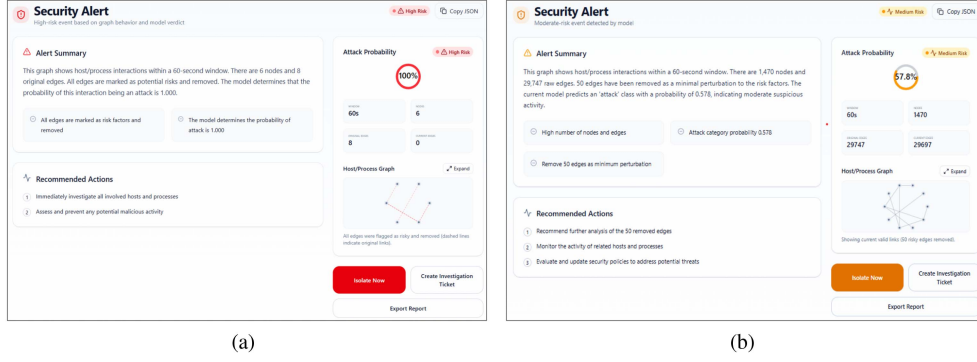


Fig. 9. Structured counterfactual summaries and recommended actions generated by the fine-tuned LLM. (a) high-risk instance; (b) medium-risk instance.

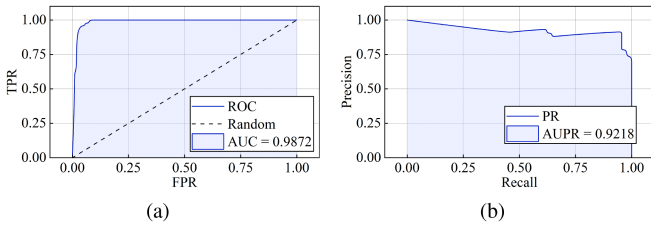


Fig. 10. ROC and PR (Precision-Recall) curve of CCG-IDS on the EVTX dataset. (a) ROC curve; (b) PR curve.

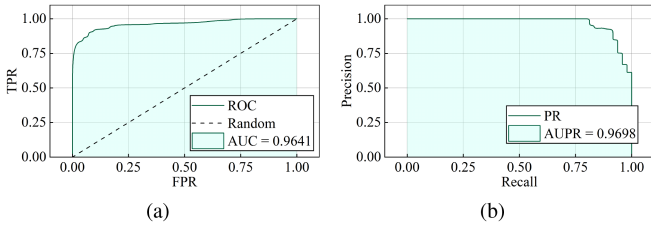


Fig. 11. ROC and PR curve of CCG-IDS on the OpTC dataset. (a) ROC curve; (b) PR curve.

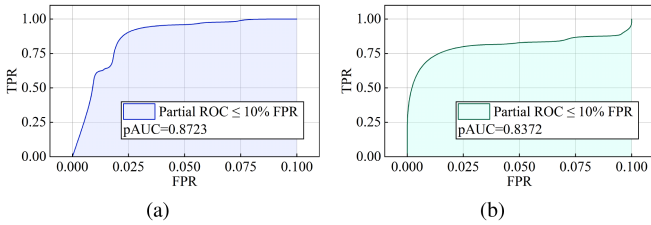


Fig. 12. Normalized partial ROC area (pAUC@10%) of CCG-IDS on the EVTX and OpTC datasets.

mean the detector sustains about 87.2% (EVTX) and 83.7% (OpTC) recall while keeping the false-positive budget under 10%. This aligns with IIoT requirements where low-FPR operation often required to avoid alarm fatigue and inadvertent process interruptions, while still maintaining high attack coverage.

Table IV compares different thresholding strategies under the same ROC curve. The RAW threshold, calibrated on the

TABLE IV
THRESHOLD COMPARISON ON THE SAME ROC.

Dataset	Method	Test FPR	Test TPR	Precision	$\alpha=5\%$
OpTC	RAW	0.0714	0.8125	0.9286	0.000283
OpTC	FISHER	0.0000	0.6208	1.0000	0.484434
EVTX	RAW	0.1028	1.0000	0.6716	0.002174
EVTX	FISHER	0.0748	0.9778	0.7333	0.003572

Thresholds are calibrated on validation at $\alpha = 5\%$ and evaluated on the test set.

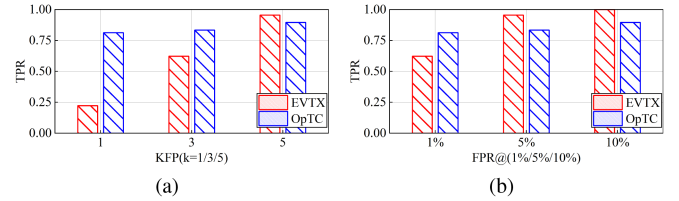


Fig. 13. Comparison of low false-positive operating points. (a) TPR at fixed FPR = 1%, 5%, 10%, interpolated. (b) TPR at fixed false-positive counts $k = 1, 3, 5$.

validation set at $\alpha = 5\%$ FPR, shifts the operating point toward a higher recall regime on test, yielding TPR = 0.8125 at FPR = 0.0714 on OpTC. In contrast, the Fisher-calibrated threshold achieves a zero-false-positive regime with FPR = 0 and TPR = 0.6208. On EVTX, RAW attains TPR = 1.000 at FPR = 0.1028, while Fisher reduces FPR to 0.0748 with only a marginal recall drop to TPR = 0.9778 and improved precision from 0.672 to 0.733. These results reveal an operational tradeoff between high-recall and zero-FP modes, offering flexible deployment choices for IIoT systems.

Fig. 13 presents recall under low false-positive constraints for EVTX and OpTC. Under a rate budget in Fig. 13(a), OpTC achieves higher TPR at FPR = 1% with 0.8125 versus 0.622. When the budget is relaxed to FPR = 5% and 10%, EVTX clearly dominates with 0.9556 and 1.000 versus 0.8333 and 0.8958. Under a count budget in Fig. 13(b), OpTC leads at $k = 1$ with 0.8125 versus 0.2222, yet EVTX overtakes at $k = 5$ with 0.9556 versus 0.8958. The two views by rate and by count are complementary and show that our method maintains controllable and strong recall across different dataset sizes and operational budgets.

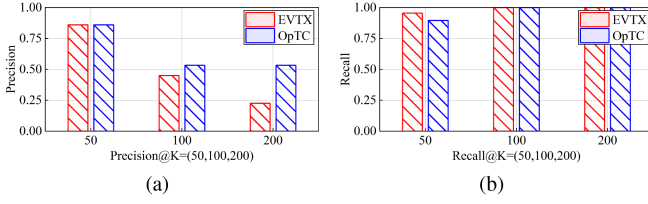


Fig. 14. Top- K alert review performance. (a) Precision at K . (b) Recall at K for EVTX and OpTC.

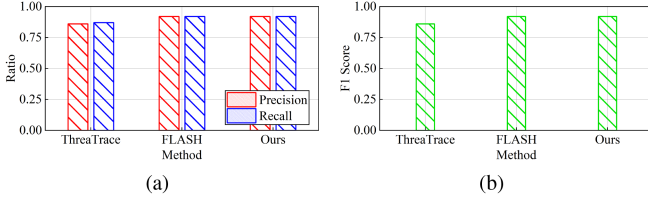


Fig. 15. Precision, recall, and $F1$ score across methods on OpTC Attack 3 (25 September). (a) Comparison of Precision and Recall; (b) Comparison of $F1$ score.

Fig. 14 shows Precision@ K and Recall@ K on EVTX and OpTC. At $K=50$, both datasets achieve high precision (0.86) with strong recall (EVTX 0.956, OpTC 0.896). Increasing to $K=100$ reaches full recall on both datasets, while precision naturally decreases as more low-score items are included. This paired view links analyst effort to expected yield and confirms that reviewing the top 50–100 alerts already captures nearly all true positives with high purity.

Fig. 15 shows the comparison of CCG-IDS against two strong baselines on OpTC Attack 3. Compared with ThreatTrace [6], CCG-IDS achieves notable gains in all metrics, improving precision by 5.6%, recall by 5.5%, and $F1$ by 6.1%, demonstrating stronger robustness and generalization on complex provenance graphs. Compared with FLASH [20], CCG-IDS attains slightly higher recall, identical $F1$ score, and identical precision, while improving the AUC from 0.95 to 0.96. This indicates that CCG-IDS achieves a more discriminative and stable decision boundary, effectively reducing overlap between benign and attack samples. Moreover, the conformal-counterfactual design of CCG-IDS provides built-in uncertainty calibration and causal interpretability, which are absent in FLASH, enabling both high accuracy and explainable threat reasoning. These results verify that CCG-IDS not only matches or surpasses state-of-the-art detection accuracy but also advances interpretability and reliability across heterogeneous attack scenarios.

C. Cross-Domain and Deployment Evaluation

1) *Cross-Domain Transfer and Threshold Portability*: To evaluate robustness under heterogeneous IIoT domains, we train the detector on a source domain and directly evaluate it on a target domain without any target fine-tuning or retraining. This setting reflects realistic onboarding scenarios where labeled attacks are unavailable in newly deployed environments. We report threshold-free ranking metrics (ROC-AUC and AUPR) as well as low-FPR operating metrics (TPR@1% FPR, TPR@5% FPR). In addition, we transfer the conformal threshold calibrated on

TABLE V
DETECTOR ABLATION ON OPTC (THRESHOLD-FREE RANKING METRICS, TEST_RAW)

Detector Variant	AUPR	AUC	TPR@1%	TPR@5%	TPR@10%
Full	0.96	0.95	0.7917	0.8333	0.8542
w/o Dual (mean)	0.90	0.87	0.5625	0.6042	0.7083
w/o Dual (scaled_sum)	0.93	0.90	0.7292	0.7292	0.7292
w/o Fisher calibration	0.95	0.93	0.7292	0.8333	0.8333
w/o DropEdge ($p=0$)	0.95	0.93	0.7708	0.7917	0.8750
Fisher scope=head	0.95	0.92	0.7292	0.7708	0.7917
Fisher scope=feature	0.90	0.89	0.1250	0.7292	0.7708

the source validation negatives at $\alpha = 5\%$ and apply it directly to the target test scores. We report the realized target FPR/TPR denoted as FPR@Thr and TPR@Thr. Table VI summarizes both directions EVTX $\rightarrow$ OpTC and OpTC $\rightarrow$ EVTX, showing that cross-domain generalization can be asymmetric and must be measured rather than assumed. The results also demonstrate that blindly reusing a numeric threshold across domains can be unsafe under score-distribution shift, motivating target-side benign recalibration for deployment.

2) *Traditional Baselines*: To broaden the comparison beyond GNNs, we include representative classical unsupervised anomaly detectors: Isolation Forest, one-class support vector machine (SVM), local outlier factor (LOF), and principal component analysis (PCA) reconstruction error. Because these methods operate on fixed-length vectors rather than graphs, we convert each time-window provenance subgraph into a lightweight window-level feature vector using simple graph summaries (e.g., node/edge counts, density) and pooled node attributes. We evaluate all methods using the same split and report AUC/AP as threshold-free ranking metrics. Table VIII shows that our detector substantially outperforms all classical baselines on OpTC in both AUC and average precision (AP).

3) *Runtime and Resource Efficiency*: We report deployment-facing efficiency measurements on the GPU. We measure detector inference latency and throughput on all windows, and separately report the explanation overhead since it is invoked only on demand for a limited number of alerts. Table VII summarizes mean and p90 latency, throughput, and peak GPU memory for both datasets. The results show that the system is practical for near-real-time deployment settings under our tested hardware configuration.

D. Ablation Study

1) *Experimental Protocol and Variants*: In Tables V–IX, detector variant means disabling or replacing exactly one module of the detector while keeping the backbone, data split, seed, and training schedule fixed. Full uses dual readout (mean+scaled-sum), and DropEdge; the Fisher module is present but only used to produce the Fisher-highU operating point as the right half of Table IX, while Table V reports threshold-free ranking on raw scores. Table V is computed from the detector’s raw scores (no Fisher-highU operating rule applied), while Fisher-highU is only used to define the conservative operating point in Table IX. In Table XI, explainer variant changes only the explanation strategy

TABLE VI
CROSS-DOMAIN GENERALIZATION AND THRESHOLD-TRANSFER STRESS TEST

Setting	AUC $\uparrow$	AP $\uparrow$	TPR@1%FPR $\uparrow$	TPR@5%FPR $\uparrow$	FPR@Thr $\downarrow$	TPR@Thr $\uparrow$
In-domain (EVTX)	0.9911	0.9290	0.6698	1.0000	—	—
In-domain (OpTC)	0.9419	0.9241	0.7500	0.7500	—	—
Transfer EVTX $\rightarrow$ OpTC	0.5079	0.5382	0.0102	0.0510	0.9524	0.9583
Transfer OpTC $\rightarrow$ EVTX	0.8113	0.7837	0.6444	0.7778	0.9673	0.9778

In-domain rows train and test within the same dataset. Transfer rows evaluate a source-trained checkpoint on the target dataset without retraining. FPR@Thr/TPR@Thr are obtained by applying the source conformal threshold ($\alpha = 5\%$ calibrated on source validation negatives) directly to target test scores.

TABLE VII
RUNTIME/LATENCY AND GPU MEMORY FOOTPRINT

Dataset	Detector (all windows)				Explainer (on-demand)		
	Lat(ms) mean	Lat(ms) p90	Throughput(g/s)	PeakMem(alloc/res)	Time(s) mean	Time(s) p90	PeakMem(alloc/res)
EVTX	1.683	1.874	399.8	0.010/0.021	0.080	0.130	0.010/0.023
OpTC	2.518	4.016	147.1	0.087/0.203	0.841	1.188	0.062/0.256

Detector metrics are per window-graph over all evaluated windows; Explainer metrics are per alerted window with greedy counterfactual search. PeakMem reports max CUDA memory allocated/reserved in GB.

TABLE VIII
COMPARED CLASSICAL (NON-GNN) ANOMALY DETECTION BASELINES WITH OURS ON OPTC

Method	AUC $\uparrow$	AP $\uparrow$
Isolation Forest	0.5283	0.5409
One-Class SVM	0.4963	0.5293
LOF	0.5362	0.5644
PCA	0.4162	0.4909
Ours	0.9641	0.9241

TABLE IX
LOW-FPR OPERATING-POINT ABLATION AT $\alpha = 5\%$ ON THE TEST SET (42 NEGATIVES, FPR = FP/42)

Detector Variant	RAW @ $\alpha=5\%$			Fisher-highU @ $\alpha=5\%$		
	FP	TPR	F1	FP	TPR	F1
Full	1	0.7917	0.8736	0	0.6875	0.8148
w/o Dual (mean)	4	0.6875	0.7765	0	0.5625	0.7200
w/o Dual (scaled_sum)	4	0.7292	0.8046	0	0.5833	0.7368
w/o Fisher calibration	2	0.8333	0.8889	N/A (Fisher disabled)		
w/o DropEdge ($p=0$)	5	0.8750	0.8842	1	0.7917	0.8736
Fisher scope=head	0	0.6042	0.7532	0	0.4792	0.6479
Fisher scope=feature	3	0.7708	0.8409	1	0.5833	0.7273

RAW is conformal threshold on raw scores. Fisher-highU is a conservative decision rule that incorporates Fisher-information-based uncertainty.

(edge ranking and search) with the detector fixed. Table X reports validation-based threshold policies that do not change model parameters.

We adopt a one-factor-at-a-time ablation design: in each experiment, only one module is disabled or replaced, while all other conditions, such as data partitioning, random seed, model backbone, and training schedule, remain identical. To explicitly reflect alarm-sensitive IIoT deployment requirements, we report two complementary metric groups: threshold-free ranking metrics (AUPR, AUC, and TPR@{1%, 5%, 10%} FPR), and operating-point metrics at a fixed low-FPR target $\alpha = 5\%$ (FP

TABLE X
VALIDATION-THRESHOLD BASELINES EVALUATED ON THE TEST SET

Threshold policy	FP	TPR	F1
val-bestF1	2	0.8333	0.8889
val-zeroFP	0	0.6875	0.8148

TABLE XI
EXPLANATION ABLATION ON ALERTED GRAPHS UNDER AN EDGE-REMOVAL BUDGET $K = 50$

Explainer Variant	$\Delta p_{\text{attack}} \uparrow$	$\frac{\Delta p_{\text{attack}}}{ E_{\text{rm}} } \uparrow$	$ E_{\text{rm}} \downarrow$	Time (s) $\downarrow$
CDS+Greedy	0.001572	4.28×10^{-5}	37.09	2.17
Random+Greedy	0.000697	1.43×10^{-5}	45.25	1.84
CDS+OneShot	0.001151	2.30×10^{-5}	50.00	1.92

We evaluate counterfactual explanations using comprehensiveness (Δp_{attack}) and sparsity ($|E_{\text{rm}}|$).

count, TPR, and $F1$). Detector ablations are summarized in Tables V and IX. Explanation-module ablations are summarized in Table XI. We additionally include validation-threshold baselines in Table X to contextualize conformal calibration.

2) Dual Readout Pooling: To test whether dual readout pooling is superior to standard single readouts, we replace dual readout with mean pooling and scaled-sum pooling, respectively, while keeping all other modules unchanged. As shown in Table V, removing dual readout reduces ranking performance (AUPR/AUC and TPR at low FPR). More importantly, at $\alpha = 5\%$ in Table IX, false positives increase from 1 to 4 (FPR: 0.0238 $\rightarrow$ 0.0952), and TPR decreases, indicating that dual readout is a key contributor to separability in the low-FPR regime.

3) DropEdge Regularization: Disabling DropEdge has a minor influence on ranking metrics but markedly increases spurious alarms at the operating point, FP increases from 1 to 5 (FPR: 0.0238 $\rightarrow$ 0.1190) at $\alpha = 5\%$ in Table IX, suggesting

TABLE XII
EXPLANATION BENCHMARK ON OPTC UNDER MATCHED EDGE BUDGET ($K = 20$), TOP- $K_{\text{ALERT}} = 20$ EXPLAINED WINDOWS, 5 REPEATS

Method	$\Delta_{\text{comp}} \uparrow$	$\Delta_{\text{suf}} \downarrow$	Ratio $\downarrow$	Jaccard $\uparrow$	Runtime(ms) $\downarrow$
CF-GNNExplainer	-0.0001 $\pm$ 0.0022	0.3079 $\pm$ 0.4555	0.00319 $\pm$ 0.01163	1.000 $\pm$ 0.000	1323 $\pm$ 180
GNNExplainer	-0.0001 $\pm$ 0.0022	0.3079 $\pm$ 0.4555	0.00319 $\pm$ 0.01163	1.000 $\pm$ 0.000	1388 $\pm$ 277
PGExplainer	0.0059 $\pm$ 0.0460	0.3374 $\pm$ 0.4610	0.00319 $\pm$ 0.01163	1.000 $\pm$ 0.000	804 $\pm$ 7977
Ours	0.0060$\pm$0.0147	0.2435$\pm$0.4518	0.00277$\pm$0.01173	0.214 $\pm$ 0.342	974$\pm$127

$\Delta_{\text{comp}} \uparrow$ and $\Delta_{\text{suf}} \downarrow$ measure faithfulness and sufficiency; Ratio $\downarrow$ measures sparsity; Jaccard $\uparrow$ measures set stability.

DropEdge mitigates incidental correlations in provenance graphs that otherwise lead to false positives.

4) *Fisher-Aware Uncertainty Calibration*: We evaluate Fisher uncertainty with/without calibration and test different uncertainty scopes (head-only versus feature-only). Table IX shows that Fisher-highU systematically shifts the decision rule toward fewer alarms and can reach the zero-FP regime (FP=0) under $\alpha = 5\%$, at the cost of reduced recall (lower TPR), which matches the design goal of conservative low-alarm IDS deployments. The scope study indicates that feature-only uncertainty is less reliable, while head-based uncertainty yields more stable low-FP control. We, therefore, adopt the default setting used in full.

5) *Conformal Thresholding Versus Validation-Based Thresholds*: To address whether conformal thresholding is preferable to simple validation-set thresholding, we add two baselines: val-bestF1 (threshold maximizing validation $F1$) and val-zeroFP (minimum threshold achieving zero FP on validation negatives). As shown in Table X, val-bestF1 tends to be more aggressive (higher FP), whereas val-zeroFP is overly conservative (lower TPR). In contrast, conformal calibration directly targets a user-specified low-FPR level at $\alpha = 5\%$ and provides finite-sample error control under standard exchangeability assumptions, yielding a principled operating point between these two heuristics in Table IX. Concretely, the conformal threshold is computed on the validation set as the $(1 - \alpha)$ -quantile of negative-sample nonconformity scores and then applied to the test set.

6) *Counterfactual Explainer Ablation*: We evaluate explanations only on alerted (high-confidence) attack predictions produced by the fixed trained detector. We ablate the explanation pipeline by varying edge ranking (CDS versus random) and search strategy (Greedy versus OneShot) under a fixed edge-deletion budget $K = 50$. Since strict label flips can be rare for high-confidence alerts when only edge deletions are allowed, we evaluate explanations using comprehensiveness and sparsity: $\Delta p_{\text{attack}} = p_{\text{attack}}(G) - p_{\text{attack}}(G \setminus E_{\text{exp}})$ and $|E_{\text{rm}}|$. Table XI shows that CDS and Greedy achieve a larger reduction in attack probability with fewer removed edges than random and greedy, and are more edge-efficient than OneShot, validating the necessity of CDS ranking and greedy acceptance for compact and faithful provenance explanations.

Attribution-based explainers assign importance scores, which identify edges correlated with the prediction but do not establish that these edges are necessary for the alarm. Our explanation is counterfactual that we search for a minimal intervention set $S \subseteq E_t$ such that removing S reduces the calibrated risk

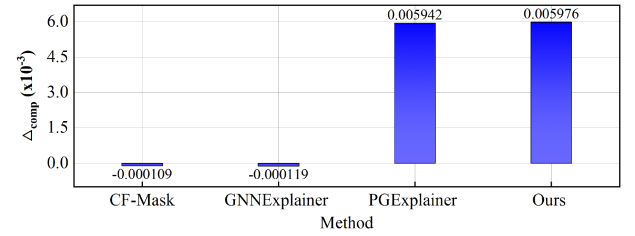


Fig. 16. Comprehensiveness comparison (Δ_{comp}) under the same edge budget $K = 20$ on OpTC. Ours yields the largest average probability drop when removing the explanation edges, indicating more decision-critical evidence.

score below the operational threshold. This produces a necessary causal chain: without these events/relations, the alert would not fire. Quantitatively, under a fixed deletion budget, we ablate the ranking and search components. CDS+Greedy achieves a larger probability reduction with fewer removed edges than random ranking and is more edge-efficient than one-shot deletion, indicating more compact and faithful decision-critical evidence.

E. Explanation Evaluation

We further provide a rigorous explanation benchmark that evaluates faithfulness, sufficiency, sparsity, stability, and runtime, and compares our counterfactual pipeline with standard graph explainers (GNNExplainer [26], PGExplainer [28], and CF-GNNExplainer [29] baseline). To rigorously validate the proposed equivalent sparsity, we repeat each method 5 times to compute stability. Given a window graph G and an explainer output edge set $E_{\text{exp}} \subseteq E$, we report the following:

- 1) comprehensiveness $\Delta_{\text{comp}} = f(G) - f(G \setminus E_{\text{exp}})$, higher is better;
- 2) sufficiency $\Delta_{\text{suf}} = f(G) - f(G_{E_{\text{exp}}})$, lower is better;
- 3) sparsity ratio $= |E_{\text{exp}}|/|E|$, lower is better;
- 4) stability as mean Jaccard overlap of E_{exp} across repeats, higher is better;
- 5) runtime per explanation (ms).

We compare with widely used explainers: GNNExplainer and PGExplainer, and a counterfactual masking baseline (CF-GNNExplainer) that is CF-GNNExplainer-like in that it optimizes an intervention mask over edges/events. Table XII reports quantitative results under the matched budget. Our counterfactual explanations achieve a larger score drop when removing the selected edges (higher Δ_{comp}) and improved sufficiency (lower Δ_{suf}) while remaining compact (low ratio), indicating more faithful decision-critical evidence. Fig. 16 visualizes the comprehensiveness comparison. Set-overlap stability (Jaccard)

can be lower for counterfactual explanations because minimal interventions are often nonunique, so that multiple distinct edge sets may suffice to cross the operating threshold. We, therefore, also report effect stability via the variance of Δ_{comp} and Δ_{suf} , showing that the explanation effect remains robust despite multiple valid minimal solutions.

V. CONCLUSION

This article proposes CCG-IDS, an interpretable graph intrusion detection system for IIoT. Methodologically, we use a unified host provenance graph as the core, homogeneously fusing windows host audit logs with public dataset events to establish a time-window-level subgraph detection paradigm. At the representation and learning level, we propose dual readout pooling to balance sensitivity to rare anomalies with subgraph stability. We also combine the joint objective of class-weighted cross-entropy and a low FPR ranking loss to directly optimize the low false-positive operating point. At the decision and threshold level, we propose a combined strategy of Fisher information-based uncertainty measures and conformal thresholds to provide transferable thresholds in zero or low false-positive modes. At the interpretability level, we design a discretized counterfactual minimum perturbation search and CDS to automatically extract key causal paths and generate structured security reports, reducing analysis costs. Experimental results show that CCG-IDS achieves an $F1$ score of approximately 92% on real-world datasets, consistently outperforming representative baselines in the low FPR range, demonstrating the superiority of our approach. In future work, we plan to extend CCG-IDS by incorporating other data modalities and evaluate its performance in larger-scale IIoT deployments.

REFERENCES

- [1] I. Makhdoom, M. Abolhasan, J. Lipman, R. P. Liu, and W. Ni, "Anatomy of threats to the Internet of Things," *IEEE Commun. Surv. Tut.*, vol. 21, no. 2, pp. 1636–1675, Second Quarter 2019.
- [2] "CISA issues emergency directive to mitigate the compromise of solarwinds orion network management products—CISA," Dec. 2020. Accessed: Oct. 23, 2025. [Online]. Available: <https://www.cisa.gov/news-events/news>
- [3] "Petya ransomware—CISA," Feb. 2018. Accessed: Oct. 23, 2025. [Online]. Available: <https://www.cisa.gov/news-events/alerts/2017/07/01/petya-ransomware>
- [4] B. Ahmad, Z. Wu, Y. Huang, and S. U. Rehman, "Enhancing the security in IIoT and IIoT networks: An intrusion detection scheme leveraging deep transfer learning," *Knowl.-Based Syst.*, vol. 305, Dec. 2024, Art. no. 112614.
- [5] M. Zipperle, F. Gottwalt, E. Chang, and T. Dillon, "Provenance-based intrusion detection systems: A survey," *ACM Comput. Surv.*, vol. 55, no. 7, pp. 1–36, Dec. 2022.
- [6] S. Wang et al., "Threatrace: Detecting and tracing host-based threats in node level through provenance graph learning," *IEEE Trans. Inf. Forensics Secur.*, vol. 17, pp. 3972–3987, 2022.
- [7] Z. Jadidi, J. Hagemann, and D. Quevedo, "Multi-step attack detection in industrial control systems using causal analysis," *Comput. Ind.*, vol. 142, Nov. 2022, Art. no. 103741.
- [8] X. Ma et al., "A comprehensive survey on graph anomaly detection with deep learning," *IEEE Trans. Knowl. data Eng.*, vol. 35, no. 12, pp. 12012–12038, Dec. 2023.
- [9] O. A. Ekle and W. Eberle, "Anomaly detection in dynamic graphs: A comprehensive survey," *ACM Trans. Knowl. Discov. Data*, vol. 18, no. 8, pp. 1–44, Jul. 2024.
- [10] T. Zhu et al., "AptShield: A stable, efficient and real-time apt detection system for linux hosts," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 6, pp. 5247–5264, Nov./Dec. 2023.
- [11] J. Wang et al., "Generalizing to unseen domains: A survey on domain generalization," *IEEE Trans. Knowl. data Eng.*, vol. 35, no. 8, pp. 8052–8072, Aug. 2023.
- [12] Z. Qiao, D. Shi, S. Jin, Y. Shi, L. Jing, and C. Qiu, "Enhancing cross-domain generalization by fusing language-guided feature remapping," *Inf. Fusion*, vol. 119, Jul. 2025, Art. no. 103029.
- [13] H. Yuan, H. Yu, S. Gui, and S. Ji, "Explainability in graph neural networks: A taxonomic survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5782–5799, May 2023.
- [14] X. Li, J. Wang, and Z. Yan, "Can graph neural networks be adequately explained? A survey," *ACM Comput. Surv.*, vol. 57, no. 5, pp. 1–36, Jan. 2025.
- [15] S. Lagraa et al., "A simple graph embedding for anomaly detection in a stream of heterogeneous labeled graphs," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107746.
- [16] F. Gao, J. Liu, C. Li, Z. Gao, and R. Zhao, "Signal-relationship-aware explainable intrusion detection in controller area networks using graph transformers," *Knowledge-Based Syst.*, vol. 328, Oct. 2025, Art. no. 114237.
- [17] D. Shan, X. Du, W. Wang, N. Wang, and A. Liu, "KPI-HGNN: Key provenance identification based on a heterogeneous graph neural network for big data access control," *Inf. Sci.*, vol. 659, Feb. 2024, Art. no. 120059.
- [18] J. Ren and R. Geng, "Provenance-based apt campaigns detection via masked graph representation learning," *Comput. Secur.*, vol. 148, Jan. 2025, Art. no. 104159.
- [19] S. He, P. He, Z. Chen, T. Yang, Y. Su, and M. R. Lyu, "A survey on automated log analysis for reliability engineering," *ACM computing surveys (CSUR)*, vol. 54, no. 6, pp. 1–37, Jul. 2021.
- [20] M. U. Rehman, H. Ahmadi, and W. U. Hassan, "Flash: A comprehensive approach to intrusion detection via provenance graph representation learning," in *Proc. IEEE Symp. Secur. Privacy*, 2024, pp. 3552–3570.
- [21] N. X. Tung et al., "Graph neural networks for next-generation-IIoT: Recent advances and open challenges," *IEEE Commun. Surveys Tut.*, vol. 28, pp. 2226–2262, 2025.
- [22] X. Han, X. Han, T. Pasquier, A. Bates, J. Mickens, and M. Seltzer, "UNICORN: Runtime provenance-based detector for advanced persistent threats," in *Proc. Netw. Distrib. Syst. Secur. Symp.* 2020, San Diego, CA, USA, Internet Society, Feb. 23–26, 2020, pp. 1–18, doi: [10.14722/ndss.2020.24046](https://doi.org/10.14722/ndss.2020.24046).
- [23] Q. Wang et al., "You are what you do: Unting stealthy malware via data provenance analysis," in *Proc. Netw. Distrib. Syst. Secur. Symp.* 2020, San Diego, CA, USA, Internet Society, Feb. 23–26, 2020, pp. 1–17, doi: [10.14722/ndss.2020.24167](https://doi.org/10.14722/ndss.2020.24167).
- [24] A. Rawal, A. Raglin, D. B. Rawat, B. M. Sadler, and J. McCoy, "Causality for trustworthy artificial intelligence: Status, challenges and perspectives," *ACM Comput. Surv.*, vol. 57, no. 6, pp. 1–30, Feb. 2025.
- [25] F. Yang, J. Xu, C. Xiong, Z. Li, and K. Zhang, "PROGRAPHER: An anomaly detection system based on provenance graph embedding," in *Proc. 32nd USENIX Secur. Symp.*, 2023, pp. 4355–4372.
- [26] Z. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: Generating explanations for graph neural networks," *Adv. Neural Inf. Process. Syst.* 32, 2019, pp. 9244–9255.
- [27] J. Tan et al., "Learning and evaluating graph neural network explanations based on counterfactual and factual reasoning," in *Proc. ACM Web Conf. 2022 ('22)*, Virtual Event (Lyon, France), Apr. 25–29, 2022, pp. 1018–1027, ACM, doi: [10.1145/3485447.3511948](https://doi.org/10.1145/3485447.3511948).
- [28] D. Luo et al., "Parameterized explainer for graph neural network," *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 19620–19631, 2020.
- [29] A. Lucic, M. A. Ter Hoeve, G. Tolomei, M. De Rijke, and F. Silvestri, "CF-GNNExplainer: Counterfactual explanations for graph neural networks," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2022, pp. 4499–4511.